Miami University Megahed

MIAMI UNIVERSITY

SIGHT

Safety Immersion and Gamified Hazard Training

FY26-27 WSIC | Q3 | February 26, 2026 Update

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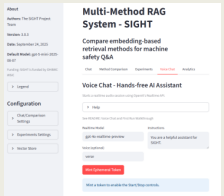
ON SCHEDULE

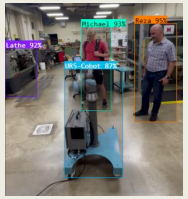
Ohio Bureau of Workers' Compensation

AI Chatbot & RAG



Multi-method RAG (v3.0.3) for machine safety Q&A with voice chat support.

Computer Vision & IoT



YOLO object detection for safety. IoT sensors for real-time monitoring.

VR Digital Experience

Browser-based CNC training with PPE, setup, and cleanup modules. Tested with 40+ users.

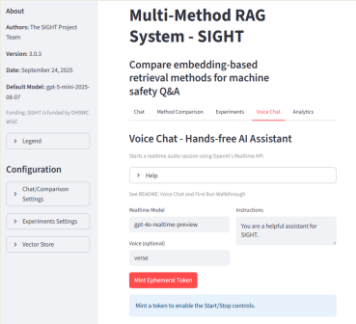
Progress Against Proposal Milestones

Phase 1	Phase 2	Phase 3	Phase 4	Phase 5
Data Collection	Core AI Dev	Gamified Training	XR Proof-of-Concept	Testing
Complete	Complete	In Progress	In Progress	Future
Ends: Oct 2025	Ends: Dec 2025	Ends: Mar 2026	Ends: Dec 2026	Ends: Jun 2027

KEY MILESTONES: AI CHATBOT, VISION & IOT

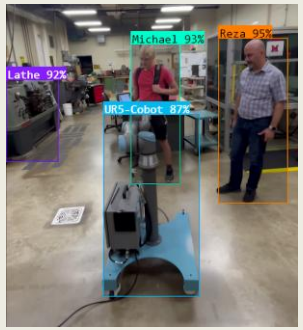
Building the intelligent infrastructure for proactive manufacturing safety

AI Chatbot + RAG System



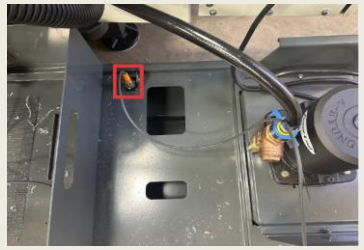
- **Tested 24 RAG pipelines across accuracy, latency, cost**
- **Top deployed pipeline: 86.66% accuracy, $0.005/query**
- **End-to-end latency averaged 10.04 seconds per response**
- **Expert-validated benchmark built for three representative machines**

Computer Vision (YOLO)

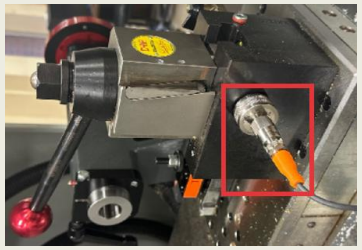


- **Motivation:**
  - OH ranks 3rd in manufacturing employment and GDP
  - Contact with objects/equipment drives 57% of injury claims
- **Identifies personnel, machines, and cobots (UR5) in real-time**
- **99%+ detection confidence across all object classes (mean average precision)**

IoT Sensor Integration



Induction Sensor



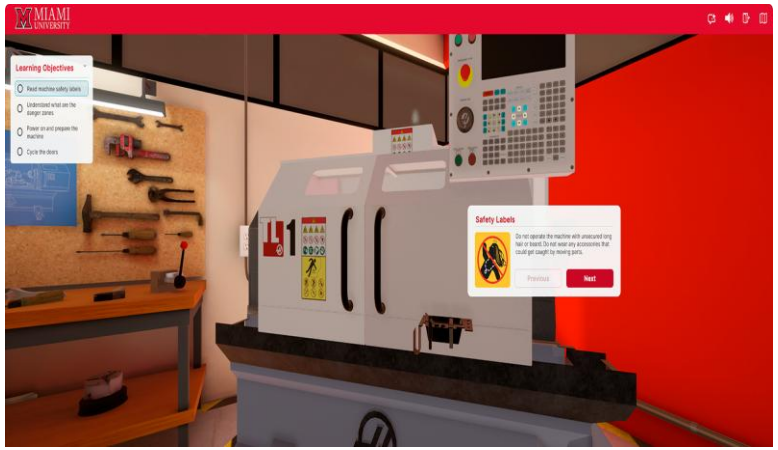
Vibration Sensor

Infrared Temperature Sensor on Milling Machine

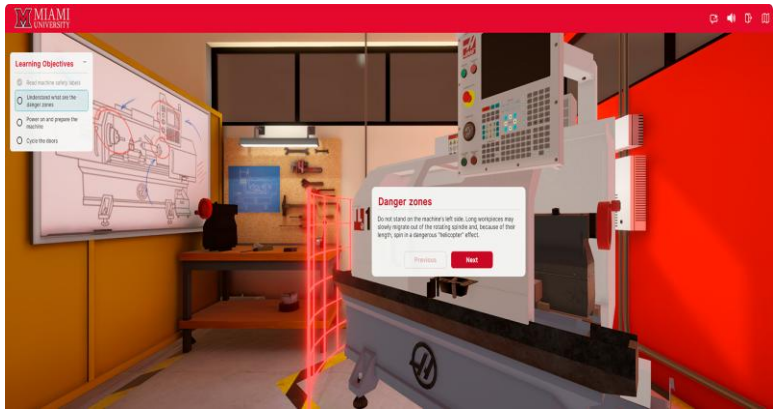
Real-time monitoring of temperature, vibration, and proximity data for safety and production/operation alerts

KEY MILESTONE: VR TRAINING EXPERIENCE PROGRESSING

Early prototype of interactive web-based CNC training deployed and user-tested with 50+ participants



Safety Labels & Warnings



Danger Zone Visualization

User Testing Feedback

51	4.2	97.5%
Testers	Avg Rating	Enjoyed or Neutral

Top Feedback Themes

- **Realistic machine interactions (air hose, chuck rotation)**
- **Clear step-by-step instructions per module**
- **PPE customization engages learners**

Machine Control Panel

PPE Training Module

Try the Live Demo

Scan or visit:

unity-auto-deploy-to-cf-page-test.pages.dev

Milling Demo

PPE Demo

UPCOMING GOALS, RISKS AND GOALS

Roadmap for the next four months and path to broader adoption

Upcoming Goals (Next 4 Months)

- **RAG:** Utilize our deployed API to connect the RAG/ chatbot to the VR and AR environments
- **VR/Analytics:** capture 90%+ of training interactions
- **IoT/AR:** Integrate and summarize real-time sensor information for AR consumption

Risks & Challenges

- RAG accuracy measured based on text-to-text interactions only
- Browser–based VR performance can vary across devices and browsers (e.g., we addressed a Firefox shader issue in unity)
- Integrating AI and CV in AR requires testing (planned)

Stakeholder Update

- **RAG/AI:** NAMRC Paper accepted with minor revisions
- Testing four **VR** training modules with 51 participants
- **VR Feedback:** Incorporated UI color choice, font scaling, and navigation improvements
- Industry reviewers appreciated the realism and guided approach of our VR training modules

Commercialization Status

- Web-based platform: zero-install access for any organization
- AI chatbot and VR modules are independently deployable
- Exploring pilot partnerships with Ohio manufacturers